

TRADE NAME: 1,3-BUTYLENE GLYCOL

First issue: 27.08.1992

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1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

SDS No.: 09-02

Trade Name:

1,3-BUTYLENE GLYCOL

Company Name: KH Neochem Co., Ltd.

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Department for Information: Sales & Marketing Department 2

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Emergency Contact Number:

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2. INFORMATION ON COMPOSITION

Chemical Name	1,3-BUTYLENE GLYCOL
Synonym	1,3-Butanediol, 1,3 BG
Content	99.5 %UP
Structural Formula	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{OH}$
Molecular Formula	$\text{C}_4\text{H}_{10}\text{O}_2$
Molecular Weight	90.1
CAS No.	107-88-0
UN No.	Not regulated.
EINECS No.	203-529-7
ENCS No.	(2)-235

3. HAZARD IDENTIFICATION

Physical Hazard:	Combustible liquid.
Major Health Hazard:	May cause drowsiness or dizziness.

4. FIRST AID MEASURES

- | | |
|-----------------------------|---|
| 4.1 General Instructions: | There are no general instructions necessary. |
| 4.2 Eye Contact: | Immediately flush eyes with plenty of clean water for at least 15 minutes while holding eyelids open. Get prompt medical attention. |
| 4.3 Skin Contact: | Remove contaminated clothing. Wash the affected area with plenty of soap and water. If irritation develops, see a physician. |
| 4.4 Ingestion: | Rinse mouth. Get prompt medical attention. Never make an unconscious person vomit or drink fluids. When vomiting occurs, keep head lower than hips to help prevent aspiration. |
| 4.5 Inhalation: | Remove individual(s) to fresh air. In emergency situations use proper respiratory protection to immediately remove the affected victim from exposure. Administer artificial respiration if breathing has stopped. Keep at rest. Get prompt medical attention. |
| 4.6 Notes to the Physician: | There is no special information available. Treat symptomatically. |

5. FIRE FIGHTING MEASURES

Flash Point (in open cup):	115 °C
Autoflammability:	377 °C
Explosion Limits:	1.9 ~ 12.6 vol%
Suitable Extinguishing Media:	Water spray, ABC dry chemicals, protein type air foams or carbon dioxide. Carbon dioxide may be ineffective on large fires, due to lack of cooling capacity and possible reignition. Not to be used for safety reasons: water jet.
Special Exposure Hazards Arising from Substance or Combustion Products:	Formation of carbon monoxide and carbon dioxide at the combustion of the product. Thermal decomposition may release toxic and irritating vapors. Thermal influence results in a pressure build-up in closed drums – danger of rupture.
Special Protective Equipment Required for Fire-fighting:	Cool product containers exposed to fire with water spray. In case of fire, wear a self-containing breathing apparatus and OSHA/MSHA approved full protective clothing. Collect contaminated water in spillage containers, if any possible and take

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care for disposal.

6. ACCIDENTAL RELEASE MEASURES

Lock all drains. Wear protective equipment (see Chapter 8.3). Remove incompatible substances. Take up product by mechanical means avoiding mist formation and place into marked, resistant container. Cover residue with an inert adsorbent, take up into resistant container and hold for waste disposal as described in Chapter 13. Carefully rinse affected ground with plenty of water.

7. HANDLING AND STORAGE

- 7.1 Handling: Handle product only in well ventilated areas. Exhaust product vapors. Check durability and applicability of tools and equipment prior to use. Exhaust released product vapors at thermal treatment directly at origin of formation. Provide eye bath and shower at working place.
- 7.2 Storage: Store product in tightly closed original containers in a dry, dark and cool place on impermeable ground. Take all measures to prevent product from entering drains or water courses in case of an accidental spill. Avoid inhalation of vapors during refilling processes. Always keep drums tightly closed.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION**8.1 Technical Facility Design:**

Install local exhaust ventilation system when treating the heated product.

8.2 Components with Exposure Limits

OSHA PELs: Not established.

ACGIH TLVs (2015): Not established.

8.3 Personal Hygiene and Protective Equipment:

Follow the general guidelines of good industrial hygiene. Avoid any direct contact with the product. Do not breathe vapor. Change contaminated clothing and thoroughly clean before reuse.

Respiratory: None required in normal use, use air supplied mask if used in confined area.

Eye Protection: Chemical safety goggles.

Hand Protection: Impermeable gloves.

Skin Protection: Work clothing and safety shoes.

Other Protective Equipment: Eye bath and washing facilities.

9. PHYSICAL AND CHEMICAL PROPERTIES

Form: Liquid.

Color: Clear, colorless.

Odor: Almost odorless.

9.1 Change in Physical State

Melting Point: -77 °C

Boiling Point: 207.5 °C

9.2 Specific Gravity: 1.006 (20/20 °C)**9.3 Vapor Pressure:** 8 Pa (= 0.06 mmHg, 20 °C)**9.4 Vapor Density (Air = 1.0):** 3.1**9.5 Solubility**

Water Solubility in H₂O: Completely soluble. (20 °C)

Solubility in Organic Solvent: Miscible with acetone and alcohol.

9.6 pH-Value: Not applicable.**9.7 Flash Point (in open cup):** 115 °C**9.8 Autoflammability:** 377 °C**9.9 Explosion Limits:** 1.9 ~ 12.6 vol%**9.10 Partition Coefficient n-Octanol/Water:** logPow = -0.29 (estimated) ¹⁾**9.11 Viscosity:** 104 mPa·s (25 °C)**9.12 Further Data:** --

10. STABILITY AND REACTIVITY

Reactivity: Stable at normal temperatures and pressure.

Conditions to Avoid: Heat, sparks and ignition sources.

Products to Avoid: Peroxides and oxidizers.

Hazardous Decomposition Products: Formation of carbon monoxide at the combustion of the product.

Polymerization: Will not polymerize.

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11. TOXICOLOGICAL INFORMATION

Acute Toxicity	oral:	rat	LD ₅₀	29.59g/kg ²⁾ , 22.8 g/kg ^{1), 2), 3)} , 12.98 g/kg ¹⁾
	dermal:	rabbit	LD ₅₀	> 20 g/kg ⁴⁾
	inhalation:	rat		No deaths from exposure to saturated vapor for 8 hours. ¹⁾
Irritation	skin:	rabbit		Not irritating. ²⁾
	eye:	rabbit		Not irritating. ³⁾
Sensitization:		human		Not sensitized. ²⁾
Mutagenicity	(in vivo Chromosome aberration assay):			Negative. ¹⁾
	(in vivo Dominant lethal assay):			Negative. (OECD TG 478) ¹⁾
Repeated Dose Toxicity	oral:	rat		Group of 10 rats were fed for 90 days on diets containing sufficient 1,3-butanediol to cause a daily ingestion of up to 5.6 g/kg without any adverse effect as judged by growth, mortality, food consumption, liver and kidney weight changes, and histopathological examination of the liver, kidney, spleen, and testes. ²⁾
Reproduction Toxicity:				Not available.
Developmental Toxicity	oral:	rat	NOAEL (6 ~ 15 D preg)	7.06 g/kg (maternal) ¹⁾ , 4.263 g/kg (offspring (F1)) ¹⁾
Carcinogenicity:	NTP			Not listed.
	IARC Monographs			Not listed.
	OSHA Regulated			Not listed.
	There were no tumors found in the 2-year feeding studies on dogs and rats. ^{1), 2)}			

12. ECOLOGICAL INFORMATION

Biodegradation:	Readily biodegradable. (79 % after 28 days, OECD TG 301C) ⁵⁾		
Bioaccumulation:	Bioaccumulation potential is low. (logPow = -0.29, estimated) ¹⁾		
Toxicity to <i>Oncorhynchus mykiss</i> :	LC ₅₀ (96 hr)	> 100 mg/L (OECD TG 203, GLP) ⁶⁾	
Toxicity to <i>Daphnia magna</i> :	EC ₅₀ (48 hr)	> 100 mg/L (OECD TG 202, GLP) ⁷⁾	
Toxicity to <i>Pseudokirchneriella subcapitata</i> :	EC ₅₀ (72 hr, growth rate)	> 1070 mg/L (OECD TG 201, GLP) ¹⁾	
	NOEC (72 hr, growth rate)	> 1070 mg/L (OECD TG 201, GLP) ¹⁾	

Do not release product into the environment. Purify polluted waste water before its release into the drains.

13. DISPOSAL CONSIDERATIONS

Comply with all federal, state and local regulations.

Do not dump this material into sewers, on the ground or into any body of water.

Preferably recover product according to official local directions, e.g. recycling, or burn the product in a chemical incinerator equipped with an afterburner and a scrubber. Empty used containers completely, rinse out thoroughly with water and dispose excluding unauthorized possible reuse.

14. TRANSPORT REGULATIONS

UN Recommendation on the Transport of Dangerous Goods

UN Class: Class not regulated.

UN No.: Not regulated.

Sea Transport IMDG CODE: No classification assigned.

Air Transport IATA/ICAO: No classification assigned.

Land Transport ADR/RID: No classification assigned.

15. REGULATORY INFORMATION

This product is not hazardous under the UN Recommendation on the Transport of Dangerous Goods.

INVENTORY Status;

UNITED STATES (TSCA): Y

CANADA (DSL): Y

EUROPEAN UNION (EINECS: 203-529-7): Y

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AUSTRALIA (AICS):	Y
JAPAN (ENCS, (2)-235):	Y
SOUTH KOREA (KE-03787):	Y
PHILIPPINE (PICCS):	Y
CHINA (IECSC):	Y
NEW ZEALAND (NZIoC):	Y
TAIWAN (TCSI):	Y

Please refer to national measures that may be relevant.

16. OTHER INFORMATION

LITERATURE:

- 1) U.S. EPA, High Production Volume Information System (HPVIS, CAS No.107-88-0)
- 2) Patty's Toxicology 5th Edition (2001)
- 3) European Chemicals Bureau, International Uniform Chemical Information Database (IUCLID, CAS 107-88-0) (2000)
- 4) NIOSH, Registry of Toxic Effects of Chemical Substances (RTECS, EK0440000)
- 5) KH Neochem Co., Ltd. Unpublished data; HODOGAYA CONTRACTLAB CO., LTD., Preliminary assessment of ready biodegradability of 1,3-BUTYLENE GLYCOL, No. D-9713 (1998)
- 6) KH Neochem Co., Ltd. Unpublished data; Harlan Laboratories Ltd, 1,3 BG: ACUTE TOXICITY TO RAINBOW TROUT (*Oncorhynchus mikiss*), PROJECT NUMBER: 0732/0176 (2009)
- 7) KH Neochem Co., Ltd. Unpublished data; Harlan Laboratories Ltd, 1,3 BG: ACUTE TOXICITY TO *DAPHNIA MAGNA*, PROJECT NUMBER: 0732/0175 (2009)

SDS Status: Revised Section 1.

Information given herein is to the best of our knowledge and is only related to the product as delivered in order to give a complete overview of safety aspects accurately as of the date indicated.
All data given are never meant to guarantee product properties.